

Towards safe control parameter tuning in distributed multi-agent systems

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Abstract— This report presents a **Decoupled Safe Bayesian Optimization (Safe BO)** architecture for tuning control parameters in safety-critical multi-agent systems. To address the optimization failures that single-Gaussian Process (GP) models suffer when handling abrupt penalty-induced gradients, we decouple the learning process into an Objective GP for performance and an independent Constraint GP for strict collision avoidance. Evaluated on a vehicle platooning simulator, our dual-GP approach successfully dampens transient string instability and guarantees zero safety violations while converging to a stable steady-state formation. Finally, a scalability analysis has been performed.

1 Introduction

Distributed multi-agent systems (MAS) are increasingly used in safety-critical real-world applications such as autonomous driving, collaborative robotics, and vehicle platooning. Vehicle platooning offers significant real-world advantages, such as road capacity, improved traffic flow, and reduced fuel consumption [1]. However, real-world deployment requires strict safety guarantees as unsafe control actions may lead to inter-vehicle collisions, instability propagation across the platoon, or dangerous behaviors in uncertain environments. The goal is therefore to optimize system performance while ensuring safe operation at all times.

Multiple agents must cooperate while making decisions based only on information available to

them and their neighboring agents, together with limited communication. That is why they are usually considered as distributed optimization problems, where each agent aims to optimize local control parameters to maximize an interdependent reward function subject to coupled safety constraints. Ensuring high performance and safety in such environments remains a major challenge, especially when the system dynamics and reward functions are unknown.

In this project, the distributed Safe Bayesian Optimization (Safe BO) framework proposed in [2] is implemented and extended by introducing an additional explicit safety constraint in the vehicle platooning scenario. In the original formulation, safety is directly coupled to the reward function through a predefined safety threshold. Consequently, a single Gaussian Process (GP) model is used both to optimize performance and to enforce safety. However, in many practical control applications, safety constraints are independent of the performance objective and must therefore be modeled separately. To address this limitation, we incorporate a second GP dedicated to modeling an additional vehicle-related safety criterion. This extension follows the constrained Safe BO formulation introduced in the single-agent setting by [3], while adapting it to the distributed multi-agent framework considered in [2]. Furthermore, a scalability analysis evaluates the operational limits of the distributed framework as the platoon expands, investigating whether the spatio-temporal Gaussian Process configuration proposed by [2] encounters computational collapse.

2 Related Literature

The optimization of control parameters in MAS has been extensively studied, particularly in safety-critical applications. Several existing approaches attempt to address this problem:

1. **Multi-Agent Reinforcement Learning (MARL):** While MARL effectively learns complex cooperative policies in unknown environments, it is notoriously sample-inefficient. In safety-critical systems like vehicle platooning, MARL’s reliance on trial-and-error exploration is often unacceptable, as it provides limited formal safety guarantees during learning and may actively sample dangerous, collision-inducing actions [2].
2. **Distributed Optimization Methods:** Techniques like the Jacobi method [4] or Alternating Direction Method of Multipliers (ADMM) [5] are commonly used to solve cost or constraints coupled optimization problems because these methods offer strong convergence guarantees. Nonetheless, they typically assume the objective functions are known or convex, which is rarely the case in black-box control parameter tuning.
3. **Distributed Bayesian Optimization** To address optimization problems involving expensive or unknown objective functions, several works have extended Bayesian Optimization (BO) to distributed and multi-agent settings. Krishnamoorthy [6] proposed decomposition-based Bayesian Optimization frameworks for large-scale distributed black-box optimization. Related works combine BO with ADMM to coordinate multiple agents while exchanging only limited local information [7]. Other approaches investigate distributed Thompson Sampling under constrained communication settings [8]. These methods significantly improve scalability and communication efficiency in decentralized systems. However, they mainly focus on optimization performance and generally do not explicitly address safety during exploration.
4. **Safe Bayesian Optimization (Safe BO):** Safe BO methods such as the SafeOpt algorithm [9] rely on GPs to model unknown

objective functions while quantifying uncertainty. By constructing confidence intervals around GP predictions, SafeOpt defines safe exploration regions and provides probabilistic safety guarantees throughout the optimization process. Extensions incorporating explicit safety constraints through additional GP models have also been proposed in the single-agent setting [3]. Despite these advances, most Safe BO methods remain centralized and assume global observability of the system state, which becomes impractical in distributed MAS due to communication limitations, scalability constraints, and privacy concerns [10].

The assigned paper [2] is unique because it addresses the problem of partial observability in a purely distributed setting. Each agent only communicates with its immediate neighbors, making the global reward function appear non-stationary from a local perspective. The paper elegantly solves this by introducing time as a latent variable [11]. By employing a custom spatio-temporal GP kernel, the algorithm allows agents to implicitly model the unobserved influence of non-neighboring agents. This enables safe and sample-efficient exploration without requiring a central authority or global information.

3 Paper Summary

The paper formulates safe controller tuning in distributed MAS as a constrained optimization problem. The objective is to maximize an unknown global reward function $f : \mathcal{A}^N \rightarrow \mathbb{R}$ while guaranteeing safety through a predefined threshold $h \in \mathbb{R}$. A MAS composed of N agents is considered, where each agent i is associated with n -dimensional control parameters $a_t^{(i)} \in \mathcal{A}_i \subset \mathbb{R}^n$, with $t \geq 1$ denoting the iteration index. For simplicity, the paper assumes identical action spaces for all agents, i.e. $\mathcal{A}_i = \mathcal{A}_j$ for all $i, j \in \{1, \dots, N\}$, and therefore writes $\mathcal{A}_i = \mathcal{A}$. The global reward function is coupled, meaning that it depends on the control parameters of all agents:

$$a_t := \left[a_t^{(1)}, \dots, a_t^{(N)} \right]^T \in \mathbb{R}^{N \times n}.$$

Safety is enforced by requiring the reward value to remain above the threshold h , leading to the

constrained optimization problem:

$$\max_{a \in \mathcal{A}^N} f(a) \quad \text{subject to} \quad f(a_t) \geq h, \forall t \geq 1.$$

Because the global reward depends on all agents, the optimization problem is strongly coupled. However, agents only communicate with their nearest neighbors, resulting in incomplete global knowledge. Consequently, identical local parameters can yield different rewards based on the actions of distant agents. These unobservable subspaces introduce severe discontinuities into each agent’s local reward function.

3.1 Safe Bayesian Optimization

The authors of the paper rely on Safe BO with GP regression, a framework known for its sample efficiency and its ability to explicitly account for safety during exploration. GPs are used to model the unknown reward function while providing both reward predictions and uncertainty estimates. However, to perform safe exploration, the paper builds upon the SafeOpt algorithm, a Safe BO method that constructs confidence intervals around the GP predictions. Based on these confidence bounds, SafeOpt defines a safe set containing actions whose lower confidence bound remains above the predefined safety threshold h . Within this safe set, the algorithm distinguishes between two types of candidate actions: potential maximizers and potential expanders. Potential maximizers correspond to safe actions that are likely to improve the current best reward estimate, thereby focusing on performance optimization. In contrast, potential expanders are safe actions associated with high uncertainty that may enlarge the currently known safe region after evaluation. The algorithm therefore balances exploitation of promising safe actions with cautious exploration of new potentially safe regions. The authors extend this SafeOpt framework to the distributed multi-agent setting by allowing each agent to maintain its own local GP model and perform safe exploration based only on locally available information. However, due to the distributed formulation, partial observability, and the introduction of temporal latent variables, the resulting safety guarantees become largely heuristic rather than strictly theoretical.

3.2 Time as a latent variable

To deal with unobservable subspaces, the authors introduce a key concept: time t as a latent variable. Instead of modeling a static reward function depending on all agents, each agent models a time-varying local reward function based only on neighboring parameters and time. The global static optimization problem is thus reformulated as a local, temporally variable optimization problem for each agent. This reformulation ensures that, at each iteration, a given observed input can correspond to only a single reward value, making GP learning well-defined. The behavior of non-neighboring agents is then implicitly extrapolated via a one-step prediction in the temporal domain.

3.3 Custom spatio-temporal kernel

To model the temporally variable reward function, the authors designed a spatio-temporal kernel separation : $k = k_S \cdot k_T$. The spatial component k_S (Matérn 5/2 kernel) captures the smoothness variation of the reward as a function of neighbor parameters. The temporal kernel k_T is designed to reflect the different phases of the optimization process. At the beginning and at the end of the optimization, the agents’ parameters evolve relatively smoothly, which is captured using an RBF kernel. However, during intermediate exploration phases, agents may abruptly change their parameters when new promising or safer regions are discovered. Such non-smooth temporal variations are better modeled by a Matérn 1/2 kernel, which allows less regular behavior than the radial basis function (RBF) kernel. A weighting function derived from Brownian motion processes is then used to smoothly interpolate between these two temporal behaviors over time, enabling the GP model to represent both smooth and abrupt temporal reward variations during distributed optimization.

3.4 Experimental validation

The algorithm is then tested using numerical examples (with 4 and 8 agents) and a simulation of heavy-vehicle (trucks) platooning. In all cases, the algorithm significantly improves the reward while respecting the safety constraints at each iteration. Comparisons demonstrate that the approach devel-

oped in the paper achieves better performance than the absence of temporal latent variables, the absence of communication, and even full communication (which is not very scalable in high dimensions, as addressed later in Section 7). A sequential expert protocol is also introduced to improve exploration and prevent it from collapsing. This protocol periodically forces neighboring agents to follow the exploration choice of a selected “expert” agent in order to improve exploration diversity and avoid premature convergence.

4 Problem Setting

In our study, we have specifically addressed the safety-critical nature of vehicle platooning, where slight controller miscalibrations can induce string instability. This phenomenon causes amplifying oscillations to cascade through the formation, ultimately leading to catastrophic rear-end collisions.

We adopt the cyber-physical platoon model of 15-ton autonomous trucks, where follower agents compute forward torque using localized proportional control subject to physical constraints. The objective is to optimize the proportional gain (K_p) to minimize steady-state error while strictly maintaining a 10-meter safety gap behind a constant-velocity leader.

To optimize the system and guarantee that safety is respected, BO using GP is highly fitting. By evaluating predictive confidence intervals, the algorithm classifies whether a new K_p parameter belongs to the verified safe set before physical execution. The system selects candidate parameters among potential maximizers and expanders, thereby guaranteeing physical safety while systematically exploring the space to maximize performance rewards.

4.1 Initial approach and its shortcomings

A fundamental limitation of the baseline architecture is its reliance on a single GP to jointly model performance and safety. By penalizing unsafe actions with severe negative scores, the system introduces sharp discontinuities into the reward

landscape. Because GPs inherently assume function smoothness, these ‘cliff-edge’ penalties conflate suboptimal performance with safety violations, ultimately degrading convergence speed and increasing the risk of collisions.

To solve this issue the performance sets and safety sets must be divided structurally [3]. We address this flaw by proposing a decoupled multi-GP architecture, which is detailed and simulated in the following section.

5 Own Approach

We have extended the assigned paper by proposing a Decoupled Safe BO architecture, along with a modified physics simulator.

5.1 Algorithmic Extension

The core limitation of the original architecture is its reliance on a single GP to evaluate both system performance and safety constraints. To resolve this, we modify the **safeopt** algorithm with the following improvements:

1. In the computational implementation of SafeOpt, the ‘cube’ refers to a discretized grid representing the bounded d -dimensional hypercube of the parameter search space, $\mathcal{X} \subset \mathbb{R}^d$. For every coordinate in this grid, the GP infers a predictive mean and variance. In our decoupled architecture, we instantiate two distinct parameter grids: **cube_objective**, which exclusively models the posterior distribution of the performance landscape (tracking error between the agents), and **cube_constraint**, which independently maps the mathematical boundaries of the safe set (minimum inter-vehicle distance). We decouple all underlying variables and data streams associated with the respective GP cubes, ensuring that physical safety and performance rewards are tracked entirely independently.
2. Inside the **acquisitionfunction**, we calculate the theoretical Safe Set (S) strictly using the constraint GP. This safe set is then forcefully overridden onto the Objective GP, mathematically restricting its search space to only safe choices.

3. By decoupling safety considerations from the objective GP, the performance score is left unbounded below. System safety is independently and strictly enforced via a `constraint_threshold` of 5.0, which dictates a minimum bumper-to-bumper clearance of 5 meters to avert collisions.

In this approach, the constraint GP first delineates a feasible parameter space mathematically guaranteed to ensure collision avoidance. Subsequently, the objective GP is strictly constrained to optimize the performance score exclusively within this verified safe set.

5.2 Physical Simulation Integration to evaluate the advantages of our extension

To interface the mathematical optimizer with the physical vehicle dynamics, we are applying substantial revisions to the `vehicle_class` file:

1. We add a wrapper class in `platoon_simulator` that intercepts the normalized agent value generated (between 0.0 and 1.0) and scales it to a realistic Proportional gain (K_p) for the vehicles between 0.0 and 2.0.
2. We are forcing the Integral gain (K_i) to 0.0, evaluating pure Proportional control phenomena. Since Integral terms induce dangerous transient velocity oscillations, eliminating K_i allows us to test whether the optimizer could successfully trade perfect distance tracking for absolute string stability and collision avoidance.

The customized physics wrapper and the decoupled Safe BO architecture are implemented in Python and the complete source code is publicly available in our Aalto University GitLab repository [12].

6 Experiments

As depicted in the cumulative reward trajectory in Figure 1, the decoupled architecture successfully maintains strict physical safety, entirely eliminating the boundary violations observed in earlier iterations.

However, an analysis of the 1D explored domains reveals a fundamental trade-off between absolute safety and global parameter exploration. While aggressive sampling previously induced severe instability and physical crashes, the strictly safe optimization yields highly localized exploration clustered near $K_p \approx 0.7$.

Although mathematically robust, the algorithm becomes highly conservative, restricting its search to a localized safe set. This tension is strictly governed by the underlying hyperparameter configurations of the two GPs.

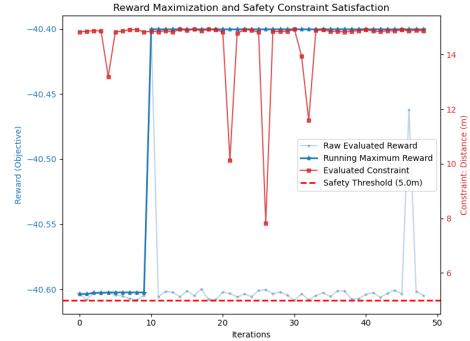


Figure 1: Reward/iteration

6.1 Kinematic Comparisons

As illustrated in Figure 3, the spatial trajectories are significantly improved in the decoupled architecture. The agents maintain better spacing, and the rubber-band effect is visibly reduced compared to the baseline. This enhancement in dynamic stability will be further confirmed in the subsequent figures.

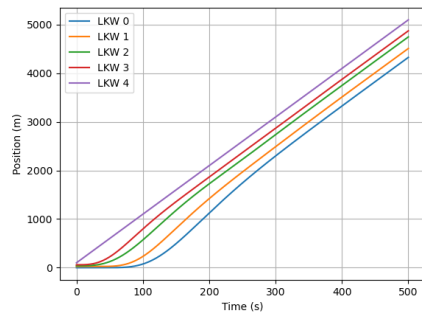
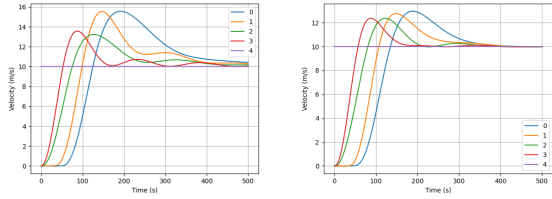


Figure 2: Decoupled Architecture Trajectory

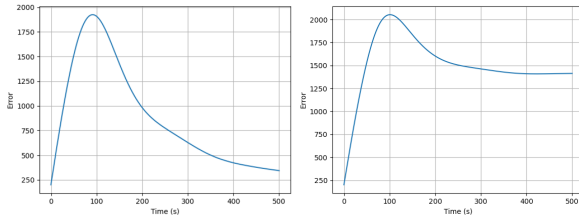
Dynamic stability is evidenced by the velocity profiles (Figure 3). The decoupled architecture (Figure 3b) significantly attenuates the peak transient velocities of trailing agents compared to the baseline, while minimizing subsequent deceleration demands. This dampening effectively suppresses the acceleration-deceleration cycles characteristic of rubber-banding, yielding a stable and safe velocity convergence for the platoon.



(a) Initial Velocity Oscillations (b) Optimized Velocity Convergence

Figure 3: Comparison of transient velocity responses

Finally, the total positional error (Figure 4) illustrates a trade-off between absolute tracking precision and string stability. The baseline model (Figure 4a) continuously drives the error toward zero, necessitating aggressive control efforts. The decoupled architecture (Figure 4b) intentionally converges to a constant, non-zero steady-state error. Prioritizing steady-state string stability over perfect tracking guarantees a rigid safety buffer, thereby eliminating the risk of rear-end collisions.



(a) Baseline System Error (b) New System Error

Figure 4: Total positional error comparisons

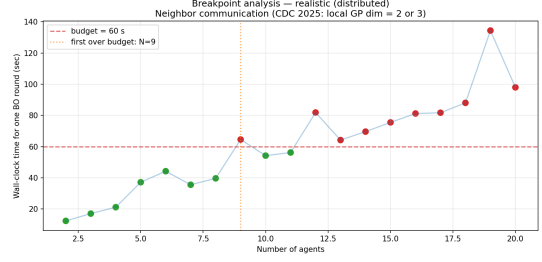


Figure 5: Wall-clock time T_{round} for one distributed BO round versus fleet size N . The dashed line is the operational budget $T_{\text{max}} = 60$ s. Markers above the line are *computationally infeasible* under that budget only

6.2 Break-point analysis

We study scalability under the same distributed setting as Tokmak et al. [2]: bidirectional nearest-neighbor communication, time as a latent variable and localized GPs that exclusively model the parameters of an agent and its direct neighbors, thereby restricting the mathematical search space to just two or three dimensions rather than the full N -vehicle fleet. With such setting the discrete SafeOpt grid does *not* suffer the dimension-induced collapse that occurs if one models all N parameters jointly. However, because our computational implementation evaluates these local models sequentially, the system still faces a linear bottleneck. We define the computational feasibility of the system based on the total wall-clock time per optimization round: $T_{\text{round}} = T_{\text{GP}} + T_{\text{platoon}}$. Since simulation time is negligible, T_{round} is bottlenecked by the GP calculations. The system is considered feasible for online learning if T_{round} does not exceed a maximum budget, T_{max} . Figure 5 shows that T_{round} grows linearly with fleet size N . Under a strict $T_{\text{max}} = 60$ s budget, the system is consistently feasible up to $N = 8$. Due to minor runtime variances, the exact breaking point fluctuates around $N = 9$ to $N = 11$. A generous $T_{\text{max}} = 180$ s budget accommodates the maximum tested size of $N = 20$.

7 Conclusion

We demonstrated that a Decoupled Safe BO architecture resolves the numerical instabilities

of single-GP models. By strictly separating the performance objective from the physical safety constraints, the system achieved zero-collision vehicle platooning. However, our results highlighted a trade-off: enforcing physical safety necessitates more conservative exploration, which limits global optimization. Besides, our scalability study showed that per-round compute time scales approximately linearly with fleet size. With a budget of $T_{\max} = 60$ s per round, configurations with $N \leq 8$ remain feasible on our hardware; larger fleets may require additional compute, parallelization, or a relaxed budget.

Future research can introduce a deeper structural analysis of the spatio-temporal kernel to better model high-dimensional interactions without over-relying on a simplified time variable. Finally, as these systems approach real-world deployment, establishing formalized, standardized mathematical protocols will be critical to providing officially verifiable safety guarantees for autonomous swarms.

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